

CH and the existence of a counterexample of norm density aleph one

[Back to Project mathematical routes](#)

Scope

The CH equivalence reuses the two continuum-density Cards and adds one direct Card. Its reverse implication applies the general density obstruction without assuming a faithful separable representation; the equivalence assumes neither CH nor its negation and makes no independence claim.

1 direct Card + 2 reused prerequisites = 3 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: CH and a Naimark counterexample of norm density aleph one

[Read this route with prerequisites](#)

Detailed route conditions

This separate route has one direct Card: the CH equivalence. It reuses the two density Cards as references; they remain members of the main route and are not duplicated as new selected Cards.

The forward implication substitutes $\mathfrak{c} = \aleph_1$ into the continuum-density existence result. The reverse applies the density obstruction to an arbitrary witness and obtains $\mathfrak{c} \leq \aleph_1 \leq \mathfrak{c}$. It does not assume that this witness has a faithful separable representation. The equivalence itself assumes neither CH nor its negation and is not a forcing or independence proof.

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (1 Card)

Level 0

Exact norm density of the fixed atomic algebra

Combines a cardinality bound from a faithful representation on a separable Hilbert space with a lower bound for every norm-dense subset.

`MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: A Naimark counterexample of continuum norm density

[Open Declaration Card](#) [Read exact source](#)

Level 1 (1 Card)

Level 1

A Naimark counterexample of continuum norm density

Gives a C^* -algebra of norm density $\mathfrak{c}$ whose nonzero irreducible representations form one unitary-equivalence class, without an identification with the compact operators.

`MathlibAnnex.CStarAlgebra.CAR.existsNaimarkCounterexampleOfDensity_continuum`

Immediate Card prerequisites: Exact norm density of the fixed atomic algebra

Used by in this scope: CH and a Naimark counterexample of norm density aleph one

[Open Declaration Card](#) [Read exact source](#)

Level 2 (1 Card)

Level 2

CH and a Naimark counterexample of norm density aleph one

Characterizes the continuum hypothesis by the existence of a Naimark counterexample of exact norm density $\aleph_1$.

`MathlibAnnex.CStarAlgebra.CAR.continuum_eq_aleph_one_iff_existsNaimarkCounterexampleOfDensity`

Immediate Card prerequisites: A Naimark counterexample of continuum norm density

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)